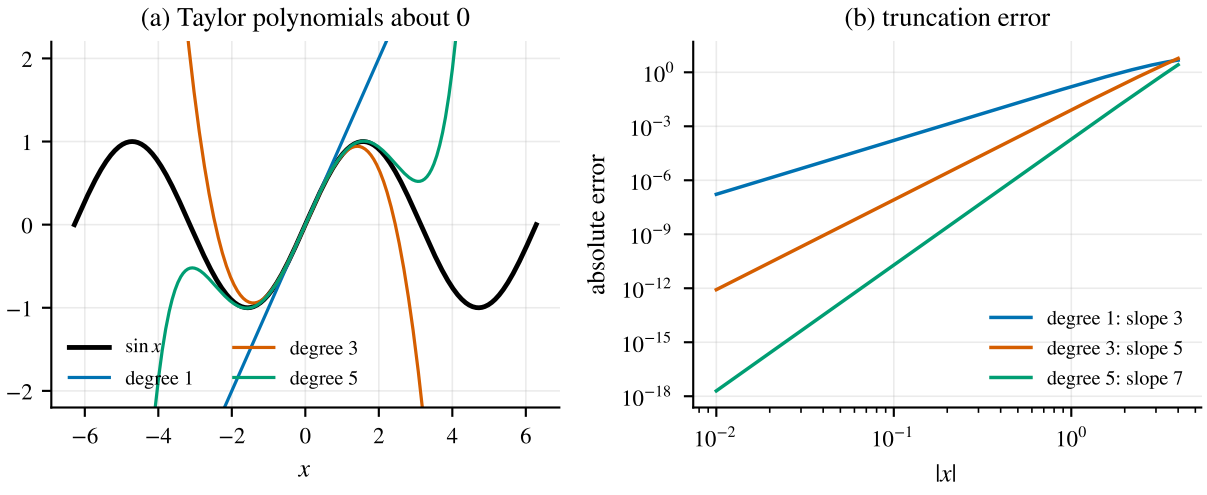
(a) Taylor polynomials about 0
2
1
0
−1
−2
−6
−4
−2
0
2
4
6
x
sin x
degree 1
degree 3
degree 5
(b) truncation error
absolute error
10^{0}
10^{-3}
10^{-6}
10^{-9}
10^{-12}
10^{-15}
10^{-18}
10^{-2}
10^{-1}
10^{0}
|x|
degree 1: slope 3
degree 3: slope 5
degree 5: slope 7